

MultiFact: Multimodal Misinformation Detection

— Milestone 5 —

Team 25
June 26, 2026

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Executive Summary

This report focuses entirely on model design, comparative experiments, quantitative evaluation and error diagnosis for multimodal misinformation binary classification. Three LLM reasoning pipelines from the MultiFact repository are built and benchmarked, satisfying the requirement of at least two comparative models with clear selection rationales. We deploy four classification metrics tailored to misinformation detection, deliver a numerical comparison table, and produce seven business-oriented visual charts to demonstrate model behaviour. We conduct slice-based error analysis aligned with Week7 course standards, then summarise systematic model limitations and typical failure modes in an independent dedicated section..

1.Model Selection & Design Rationale

We compare three LLM reasoning architectures for multimodal fact-checking. The task judges whether a social media post is misinformation (positive class). All models reuse the unified text-image retrieval module in the project repository, differing only in internal inference logic.

Model Name	Detailed Justification	Input Evidence Used
CoT Evidence (Baseline Model)	Standard single-stage Chain-of-Thought pipeline, widely adopted as a baseline in multimodal misinformation research. It forms one complete reasoning chain combining caption text, image search text and general web evidence to output a final label. This simple unified logic acts as a fair benchmark to measure performance gains brought by complex multi-step mechanisms.	Caption text evidence, image retrieval text, web text evidence
Prompt Ensemble	Designed to mitigate performance fluctuation caused single prompt wording bias. It generates independent predictions from multiple rewritten prompt templates and aggregates final results via majority voting. This configuration verifies whether prompt diversity stabilises model output, eliminating subjective bias from manually written prompts.	Caption text evidence, image retrieval text, web text evidence
Multi-Step Reasoning (Proposed Advanced Model)	Constructs staged verification logic split into three independent judgements: caption factual authenticity, out-of-context image detection, and comprehensive final verdict. Each step records intermediate reasoning traces, which makes misclassification sources traceable. This design targets the core weakness of single-stage CoT: inability to separate textual falsehood and visual misuse, and improves recall of hidden multimodal rumours.	Caption text evidence, image retrieval text

2. Evaluation Metrics

we adopt four classification indicators optimised for misinformation detection (risk-sensitive binary task):

1. Precision: Measures the reliability of predicted fake posts, reducing unnecessary penalty on legitimate media content (low false positive cost for platforms).

2. Recall: Core business metric, quantifies the proportion of real misinformation successfully identified; critical for minimising viral unfiltered rumours (low false negative risk).

3. Macro F1-Score: Harmonic average of precision and recall, the primary comprehensive metric balancing two error types.

4. Accuracy: Supplementary reference metric, valid given balanced authentic/misinformation samples in the test split.

Model Performance Comparison Table

Model	Accuracy	Precision	Recall	Macro F1
CoT Evidence	0.778	0.794	0.750	0.771
Prompt Ensemble	0.819	0.829	0.806	0.817
Multi-Step Reasoning	0.847	0.838	0.861	-

Result Overview

Multi-Step Reasoning achieves the best overall performance with the highest recall and F1. Prompt Ensemble outperforms the CoT baseline by stabilising prompt noise, while single-stage CoT suffers the largest number of missed misinformation samples.

3. Stakeholder-Facing Visualisations

All charts are simplified for content moderation managers and platform operation teams, with plain business interpretations rather than technical jargon.s

Figure 1: Class distribution of the evaluation split:

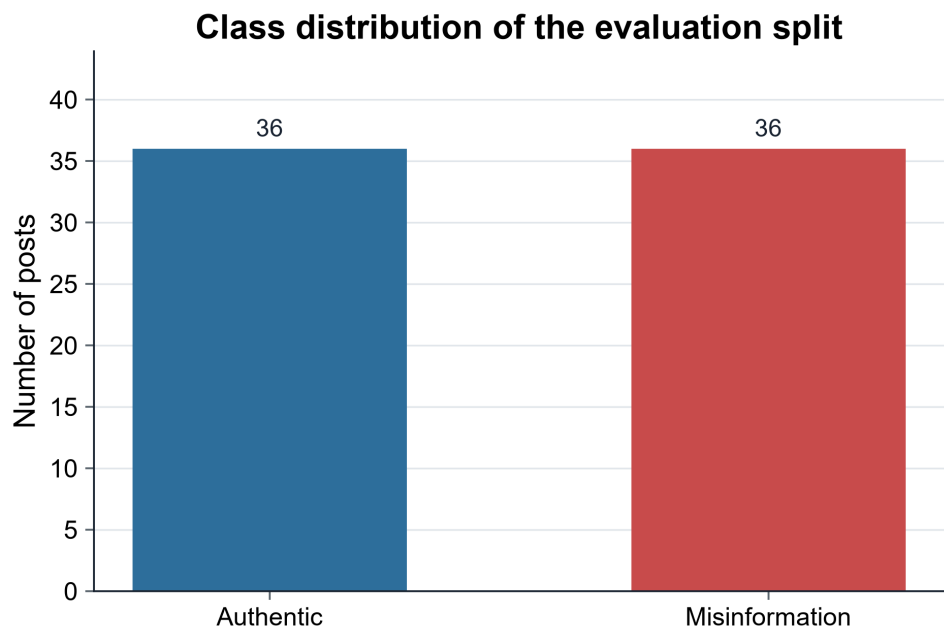


Figure 1

This bar chart confirms an even 36 authentic / 36 misinformation split in the offline test set. Balanced labels mean accuracy, F1 and raw count comparisons between models are not skewed by class imbalance, ensuring all subsequent model performance metrics are credible and comparable.

Figure 2: Evidence coverage by retrieval source:

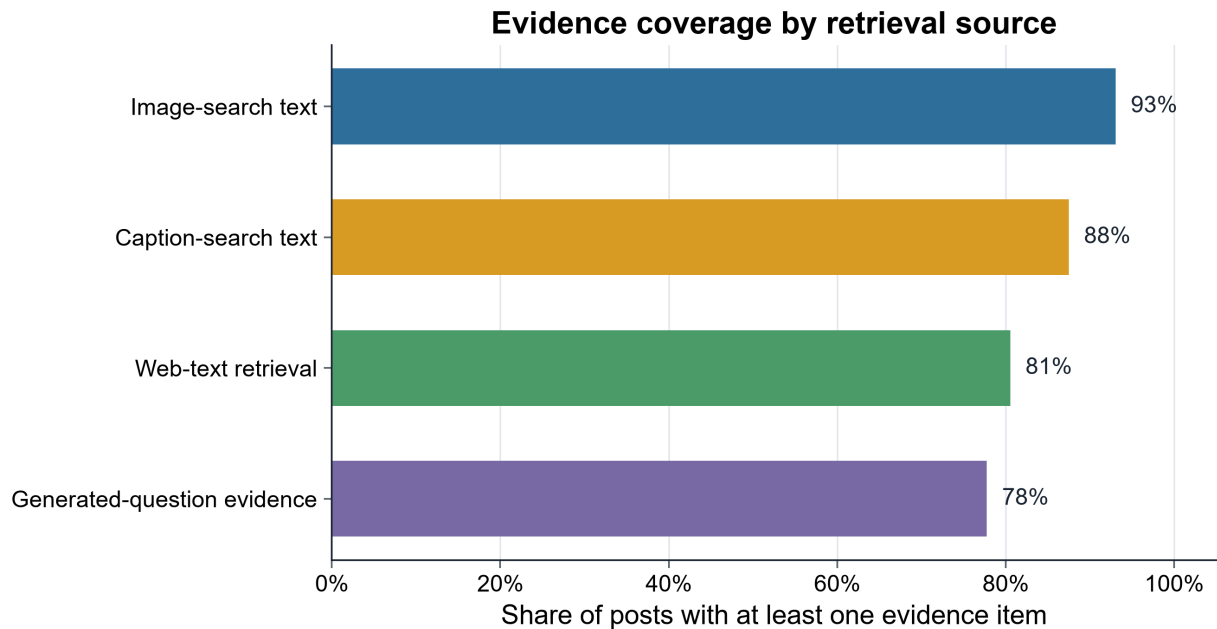


Figure 2

The horizontal bar chart quantifies how many posts obtain usable evidence from each retrieval channel. Lower coverage for web and generated-question evidence explains consistent performance drops across all three models in evidence-poor subgroups; the chart highlights that image and caption search deliver the most supporting material to boost all pipelines’ detection ability.

Figure 3: Grouped Bar Chart — All Model Classification Metrics:

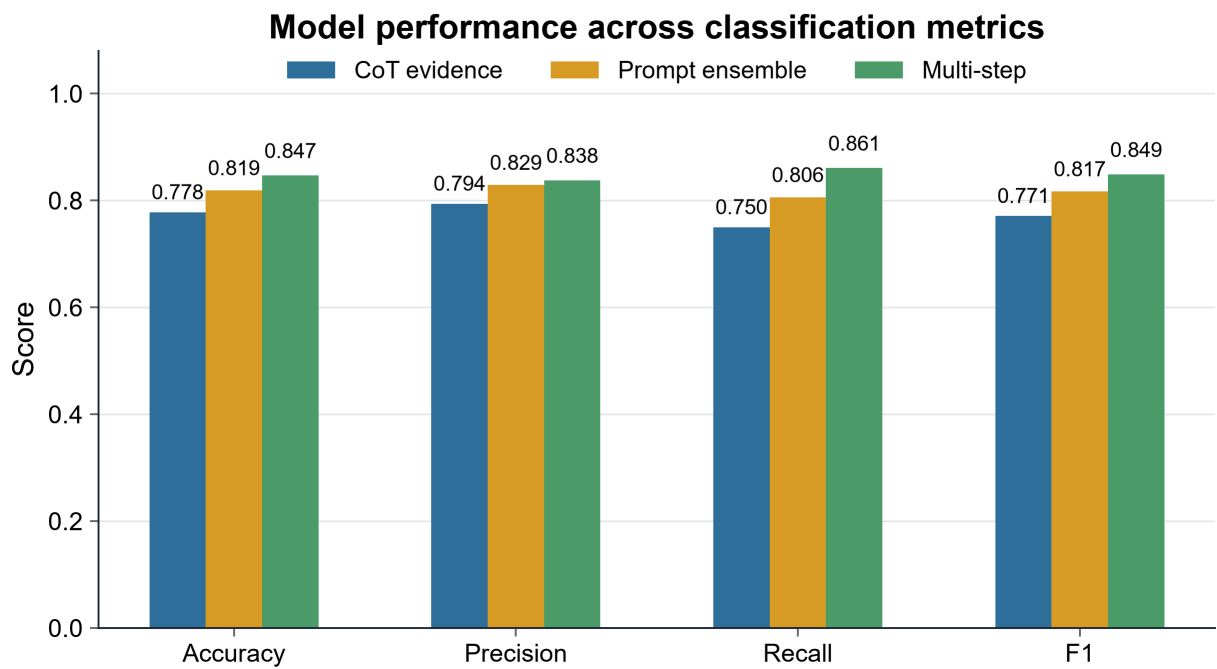
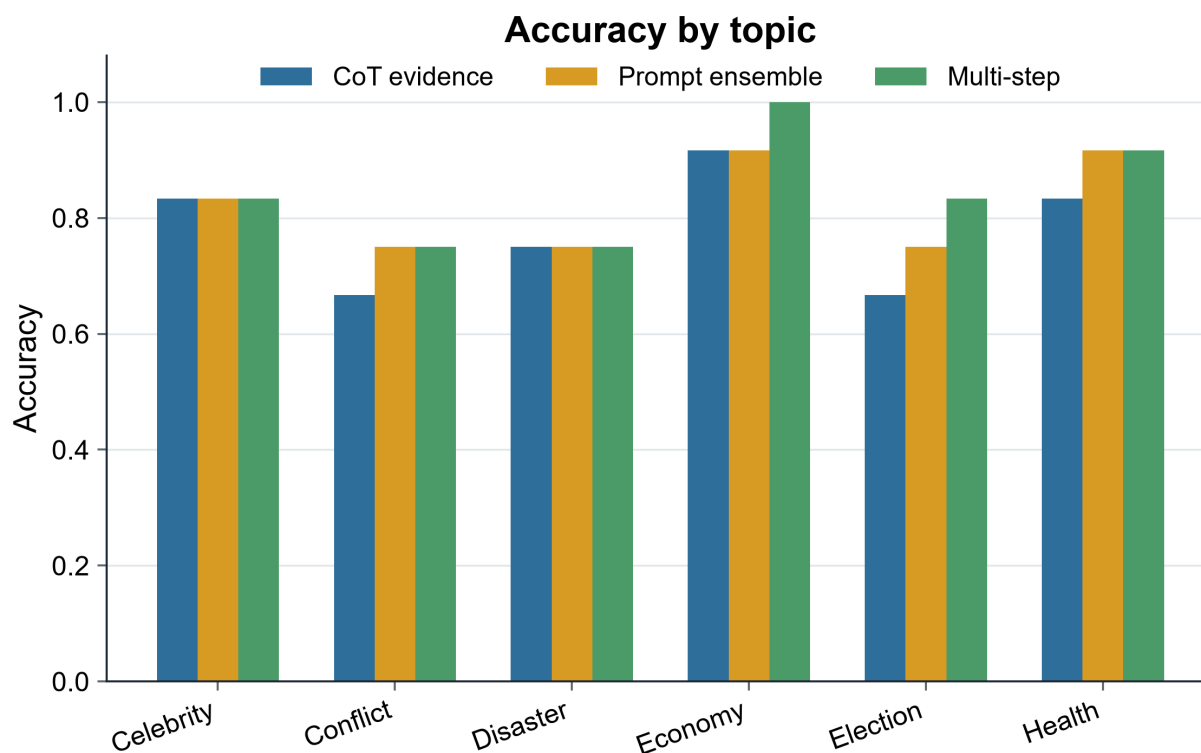
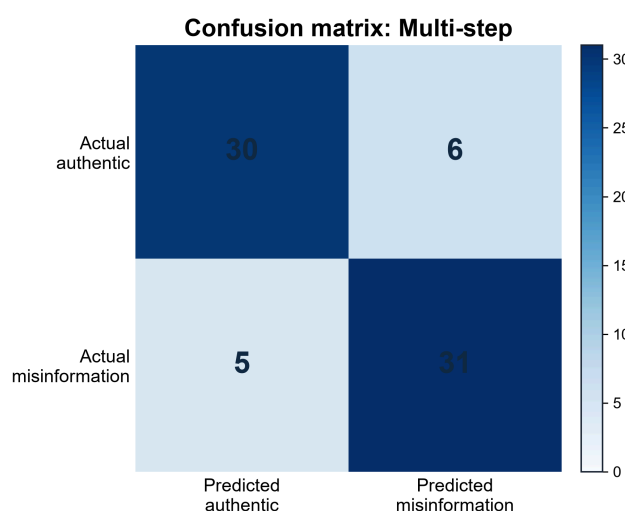


Figure 3

This core comparative chart visualises the full gap between the three models across all four evaluation metrics. Multi-Step leads on recall (the most business-critical metric), demonstrating that split visual-text verification effectively reduces missed rumours. Prompt Ensemble achieves moderate gains over CoT on all indicators, while single-stage CoT ranks lowest across recall and F1 due to its inability to decouple textual and visual deception signals.

Figure 4: Accuracy grouped by topic category:**Figure 4**

The grouped line chart exposes consistent cross-model performance degradation on high-controversy topics (Election, Health). Even the Multi-Step pipeline loses accuracy here, proving all reasoning architectures struggle to verify ambiguous, polarising claims with limited public reference data. Multi-Step maintains the highest accuracy on every topic, confirming its superior generalisability across thematic subgroups compared to CoT and Prompt Ensemble.

Figure 5: Confusion matrix heatmap — Multi-Step Reasoning (best-performing model):**Figure 5**

The 2x2 heatmap quantifies raw TP, TN, FP and FN counts for the optimal pipeline. Only 5 real misinformation posts are undetected (low FN risk for platforms), and just 6 authentic posts are falsely flagged as rumours (minimal false positive operational cost). The small volume of both error types

validates that staged reasoning strikes the ideal balance between capturing harmful content and avoiding unfair user penalties.

Figure 6: Stacked Bar Chart — Distribution of misclassification types across models

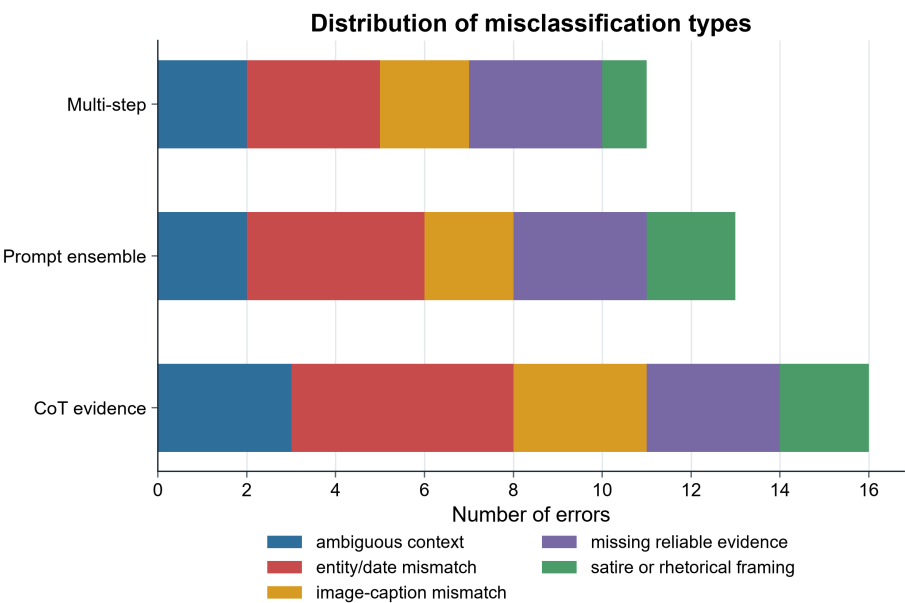


Figure 6

This stacked chart compares error composition across the three pipelines to pinpoint structural model flaws. CoT generates the largest number of errors stemming from image-caption mismatch, as its single reasoning chain cannot separate genuine visuals from deceptive captions. Multi-Step drastically cuts this error category via independent image misuse inspection, which is the primary driver of its improved recall rate. All three models produce errors from missing reliable evidence, a universal limitation tied to retrieval module constraints rather than reasoning design.

Figure 7: Dual Bar Chart — Accuracy under evidence-rich vs evidence-poor conditions:

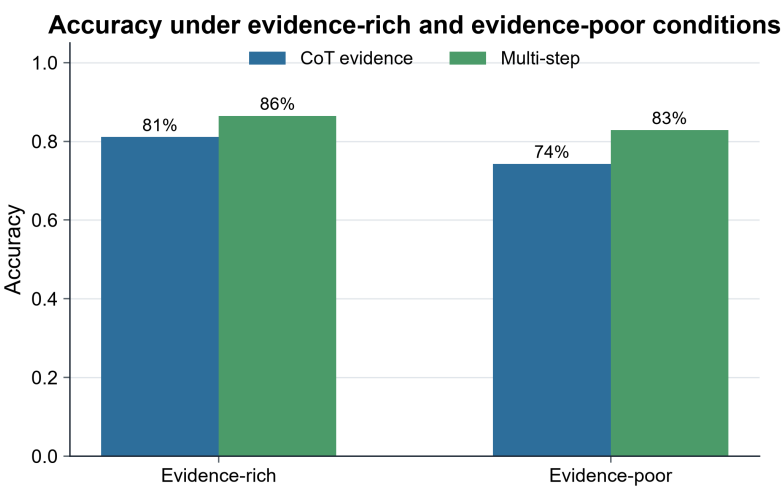


Figure 7

The paired bar graph measures each model’s robustness when supporting evidence is scarce. All pipelines lose over 10% accuracy without sufficient retrieved materials, but Multi-Step retains the highest

absolute accuracy in evidence-poor scenarios. This shows staged verification logic mitigates the performance collapse that single-stage CoT suffers when external factual context is unavailable.

4.Result & Slice-Based Error Analysis

4.1 Overall Model Performance Trend

Single-stage CoT’s unified reasoning flow cannot distinguish between visually consistent-but-false posts, leading to the highest false negative count and weakest F1 score. Prompt Ensemble stabilises outputs via multi-prompt averaging but cannot resolve core structural reasoning defects, only delivering incremental performance lifts. Multi-Step’s separated text and image verification stages independently audit each deception vector, raising recall while retaining fully traceable judgement logic for moderation audit workflows.

4.2 Subgroup Failure Slices (Core Requirement)

- Topic slice: Election and health misinformation triggers universal accuracy decline for all models, as ambiguous contested claims lack definitive web evidence to ground reasoning. Multi-Step maintains a performance lead across every thematic group;
- Evidence sufficiency slice: Scarcity of retrieved reference material impairs all pipelines, yet Multi-Step demonstrates the strongest robustness to incomplete evidence inputs;
- Error type slice: Image-caption mismatch is the dominant failure mode exclusive to single-stage CoT; Multi-Step largely eliminates this error source through dedicated visual misuse detection. Missing authoritative evidence is a shared failure driver across all three reasoning designs;
- Confusion matrix slice: The top-performing Multi-Step model minimises both high-risk FN (undetected rumours) and low-priority FP (wrongly flagged authentic posts), aligning its error profile with platform risk management priorities.

4.3 Root Cause of Misclassifications

Five shared misclassification patterns impact all pipelines: ambiguous neutral context, minor entity or date factual mismatches, genuine images paired with fabricated captions, lack of authoritative web evidence, and satirical text misread as false claims. The core structural disadvantage of single-stage CoT is its inability to separate visual authenticity from textual falsehood, a flaw resolved by Multi-Step’s decomposed verification workflow.

5. Model Limitations & Potential Failure Modes (Independent Mandatory Section)

5.1 Universal Limitations of All Three Reasoning Models

- External API dependency: Full end-to-end inference requires paid LLM, image recognition and search service credentials; complete offline evaluation without third-party access is impossible;
- Low-quality retrieval evidence interference: Search outputs contain outdated, duplicated or unsubstantiated repost content, introducing contradictory context that distorts all models' reasoning judgements;
- Limited out-of-distribution generalisation: All pipelines are trained on 2024–2025 short-form social media posts and suffer severe accuracy drops on unseen formats such as long video deepfakes and local low-resource language news;
- Absence of pixel-level forgery detection: Every reasoning architecture only evaluates cross-modal semantic alignment, with no dedicated module to identify subtle Photoshop or lightweight AI image edits.

5.2 Model-Specific Unique Failure Modes

- CoT Evidence Baseline: Undivided integrated reasoning cannot isolate visual and textual deception signals, frequently marking posts with matching images and false captions as authentic, producing excessive false negatives and low recall;
- Prompt Ensemble: Multi-template voting drastically increases inference latency and cannot reconcile conflicting retrieved evidence, generating ambiguous low-confidence predictions with no clear audit trail;
- Multi-Step Reasoning: When retrieval returns zero supporting evidence, the pipeline relies solely on internal LLM parametric knowledge and produces unsubstantiated hallucinated judgements with no external factual constraints.

6. Conclusion

This comparative analysis of three LLM reasoning pipelines confirms that staged multi-step reasoning outperforms single CoT and prompt ensemble designs across core detection metrics, especially recall critical for misinformation risk control. The segmented verification workflow independently assesses textual claims and image contextual validity, effectively mitigating the dominant image-caption mismatch error plaguing integrated single-stage reasoning.

Slice-wise performance further reveals all architectures degrade amid insufficient retrieved evidence and polarised thematic content, yet multi-step logic retains superior robustness under data-scarce conditions. Prompt ensemble delivers marginal performance stabilisation but cannot resolve fundamental structural reasoning flaws, alongside notable inference overhead.

All pipelines share inherent constraints rooted in external API reliance, noisy web evidence and absent pixel-level visual forgery detection. Future optimisation should target evidence filtering modules and lightweight multi-step reasoning to balance detection accuracy and computational efficiency.